COASTS

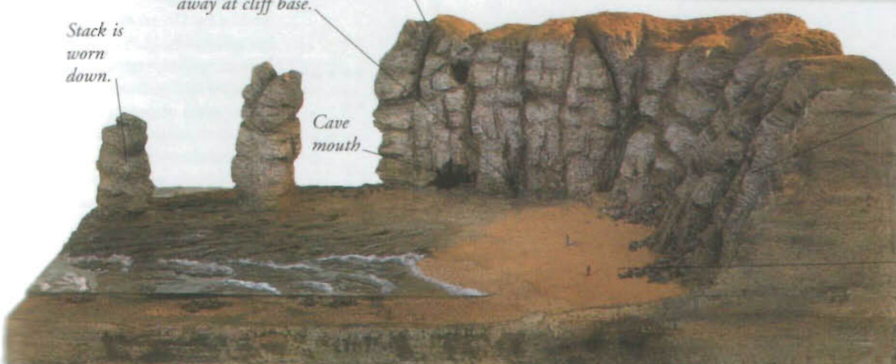
A COAST IS SIMPLY defined as the boundary between the land and sea – an area that may range from a rocky cliff to a sandy beach. This boundary is always shifting as the sea continues its relentless assault on the land – waves roll up and down, and tides ebb and flow. The action of the sea creates distinctive landforms, such as a cliff, created by eroding (wearing away) rock; a shore (an area between low tide levels and the highest storm waves); or a beach, built up by shore deposits. Wind and rain erosion also contribute to the changing aspect of coastlines.

Evolution of a coast

Waves crash against a shore with great force, wearing away rocks by pounding them with water, and hurling rocks and stones at them. On high coasts, the waves undercut the foot of the slope, creating a cliff. The model below shows the gradual effect of waves and seawater on the coast.

Beaches and sandbars

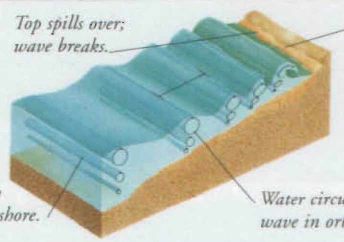
Material worn away from rocky coasts is pounded by waves into sand and shingle and deposited elsewhere as beaches and sandbars – an offshore strip of sand or shingle. A spit resembles a sandbar, with one end attached to the land; a tombolo is a spit that links an island to the mainland.



Waves

The wind whips the sea's surface into waves. Waves travel across the water, but the water in them circulates on the spot. When waves reach the shore, the bottom touches the beach and slows down; the top spills on, causing the wave to "break".

Wave formation



Coastal protection

When waves strike a beach, they wash sand or pebbles across the beach at an angle,. This repeated process is known as longshore drift. Fences or groynes may be built, to slow down such reshaping of the beach.

Coastal fences

Beach material

Fine sand and silt are usually found lower down a beach; bigger storm waves wash gravel and pebbles higher up. On some beaches, there is a ridge of pebbles, called a storm beach, which has been flung up beyond the high-tide mark by violent storms.

Types of coasts

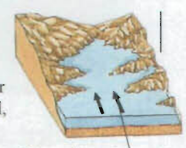
Coasts vary according to their composition and structure. Whether the coast is high or low, and made of soft or hard rock, affects whether it has been formed largely as a result of erosion or by deposition.

Bay-head beach

This is formed when material eroded from headlands (high land jutting into the sea) is washed into a bay, a coastal inlet between the headlands.

Drowned coast

Where the sea-level has risen or the land sunk, valleys are flooded to form narrow inlets, or rias. Where the valleys are glacial, the inlets are called fjords.



Highland coast

Where the sea meets a highland coast, it generally wears away the rocks, creating cliffs, small coves, and wave-cut rock platforms.

Lowland coast

Broad beaches, salt marshes, and estuaries are features of lowland coasts.

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